

ABHRANKAN CHAKRABARTI

Mathematics • Algorithms • Programming • Computational Science

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EDUCATION

Bachelor of Technology – Computer Science & Engineering

Techno Bengal Institute of Technology, Kolkata (MAKAUT)

CGPA: 7.69 (Till 5th Semester)

2023 – 2027 (Expected)

Sem 1	Sem 2	Sem 3	Sem 4	Sem 5
7.63	8.00	7.73	7.33	7.75

TECHNICAL SKILLS

Languages: C, C++, Java, Python, TypeScript, JavaScript

Web & Backend: HTML, CSS, React (Vite), Node.js, Express.js, REST APIs

Databases: MongoDB (aggregation pipelines, ACID transactions)

AI / ML Libraries: NumPy, Pandas, scikit-learn

AI / ML APIs & Inference: Gemini Pro Vision, DALL-E 3, Stable Diffusion XL, llama.cpp

Core Areas: Algorithms, Number Theory, Numerical Methods, Cryptography, Computational Mathematics

Tools & Practices: Git, GitHub, Linux, VS Code, Jest, ESLint, MVC Pattern, Dependency Injection

PROJECTS

Almost Isosceles Pythagorean Triples Optimizer

github.com/Abhrankan-Chakrabarti/almost-isosceles-pythagorean-triples

Efficient generation and optimization of almost-isosceles Pythagorean triples using number-theoretic insights.

Published an accompanying article on the mathematical exploration.

Catalan π Algorithm

github.com/Abhrankan-Chakrabarti/pi-calculation-with-catalan-numbers

Mathematical exploration of π computation using Catalan-number-based formulations and high-precision numerical optimization techniques.

Cryptify – Encryption Tool

github.com/Abhrankan-Chakrabarti/Cryptify

A cryptography-focused project implementing advanced encryption algorithms, exploring secure computation design and cybersecurity principles.

VisionaryAI (GeminiFusion)

github.com/Abhrankan-Chakrabarti/GeminiFusion

Web application leveraging Gemini Flash, DALL-E 3, and Stable Diffusion XL for chatbot interaction, image captioning, and AI image generation. (6 stars on GitHub)

Interactive Graphing Calculator

github.com/Abhrankan-Chakrabarti/GraphingCalculator

Dynamic graphing and visualization tool for mathematical functions and computational analysis.

Complex Calculator (C++ & Python)

github.com/Abhrankan-Chakrabarti

Menu-driven modular calculator supporting complex mathematical operations, implemented in both C++ and Python.

LlamaInteract

github.com/Abhrankan-Chakrabarti/LlamaInteract

Local large language model interaction system built using llama.cpp for efficient on-device inference, enabling private and resource-conscious LLM usage.

Prime Grid Visualizer

github.com/Abhrankan-Chakrabarti/prime-grid-visualizer

Fast, interactive browser-based tool to explore and visualize prime numbers. (5 stars on GitHub)

EXPERIENCE

Winternship 2025 – Full Stack Development Trainee

2025

Ajrasakha Hackathon / Online Program

Remote

- Completed structured case studies across the full MERN stack: TypeScript, React (Vite), MongoDB, and Express.js.
- Built and deployed two live React apps (Task Manager, Component App) hosted on Render.
- Implemented ACID-compliant MongoDB transactions, aggregation pipelines, and CRUD workflows.
- Developed modular REST APIs in Express with middleware chains, MVC pattern, and dependency injection.
- Applied design patterns (Observer, Strategy), generics, and advanced TypeScript types across 19+ lessons.

ARTICLES & WRITING

- Calculating π with Catalan Numbers
- Almost Isosceles Pythagorean Triples
- The Golden Ratio
- History of Cryptography

Published on personal blog: abhrankan.dpdns.org

CERTIFICATIONS

NPTEL – The Joy of Computing Using Python | Elite + Gold | Score: 91% | Top 1% (Topper)

IIT Madras / Swayam NPTEL certification (Jan–Apr 2024, 12-week course). Ranked in the Top 1% among 12,432 certified candidates. Online Assignments: 24.06/25 | Proctored Exam: 66.97/75 | Roll No: NPTEL24CS57S559700017

AREAS OF INTEREST

Artificial Intelligence • Mathematical Computing • Algorithms & Optimization • Number Theory • Cryptography • Scientific Computing • Numerical Methods

HOBBIES & ACTIVITIES

Coding • Solving mathematical problems • Learning new technologies • Recitation • Singing